DS 574 **Behavioral Models Worksheet**
Fall 2026 Prof. Kira Goldner

Behavioral Models

Sometimes, people might come across as irrational, but it can really be explained by varying attitudes toward money or probabilities. Some people prefer certainty to uncertainty. Some people value a dollar less when they already have many dollars. This is the topic we'll cover today.

Which Gamble?

	Gamble A	Gamble B
Original choice	\$10 for sure	\$20 with probability 1/2; \$0 otherwise
Add \$990	\$1000 for sure	\$1010 with probability 1/2; \$990 otherwise

Which gamble do you prefer in each row? What if the \$20 payment were instead \$50 or \$200 (so the corresponding payment in the second row becomes \$1040 or \$1190)? Does adding \$990 to every outcome change your preference?

Expected utility. Let w be the money we start with, and let m describe how much we value the amount we end up with. A gamble adds x dollars with probability q and y dollars with probability $1-q$. Its expected utility is

$$q\,m(w+x) + (1-q)\,m(w+y).$$

In the \$20-or-\$0 gamble, $x = 20$, $y = 0$, and $q = 1 - q = 1/2$; $w + x$ and $w + y$ are the resulting amounts of money. For example, $m(z) = \sqrt{z}$ for $z \geq 0$ has decreasing marginal value for money.

Small and Large Stakes

Consider two fair coin flips. Declining either gamble leaves wealth unchanged.

	Heads	Tails
Small gamble	Gain \$11	Lose \$10
Large gamble	Gain \$200	Lose \$100

Would you take each gamble? What if the large gamble offered a gain of \$1000 or \$10,000, still with a possible loss of \$100? Could a large enough gain make it attractive?

A calibration puzzle. Rabin's calibration result says: if an increasing, concave utility function over total wealth rejects the small gamble at *every initial wealth level*, it also rejects the large gamble for every finite possible gain. The assumption concerns choices across wealth levels, not one rejection at one wealth level. This challenges explaining persistent small-stakes risk aversion solely by diminishing marginal utility of wealth.

The Allais Paradox

Choose between A and B, then between C and D.

- **First choice.** A: \$1 million for sure.
  B: a 10% chance of \$5 million, an 89% chance of \$1 million, and a 1% chance of nothing.
- **Second choice.** C: an 11% chance of \$1 million; otherwise nothing.
  D: a 10% chance of \$5 million; otherwise nothing.

These departures from expected utility motivate cumulative prospect theory [**?**]. We will focus on loss aversion, one part of that broader theory.

Loss Aversion

For now, assume $m(x) = x$.

The small-stakes puzzle suggests another possibility: the agent evaluates *changes* from a reference point. A *reference point* is the position used to decide what counts as a gain or loss; here it is the status quo. *Loss aversion* means that losses weigh more heavily than gains of the same size.

Let x be a change from the reference point, measured in dollars. We use a simple value function $\phi(x)$: a gain has weight 1, and a loss has weight $\lambda \geq 1$. The agent averages $\phi(x)$ using the actual outcome probabilities and compares with $\phi(0) = 0$ for no change.
Complete the loss side of the function and sketch it for $\lambda = 2$.

$$\phi(x) = \begin{cases} x & \text{if } x \geq 0, \\ \underline{\qquad\qquad} & \text{if } x < 0. \end{cases}$$

What happens when $\lambda = 1$?

How much gain compensates for a loss? Set $\lambda = 2$. Return to the fair coin flips: gain \$11 or lose \$10; gain \$Y or lose \$100. Does the agent accept the first? What is the smallest Y for which the second is at least as good as declining?

For a fair gamble gaining $g > 0$ or losing $\ell > 0$, give the acceptance condition in terms of λ.

The Endowment Effect

Buying versus selling. Consider the same person with the same underlying mug value $v = 10$ and loss multiplier $\lambda = 2$. Measure a change of one mug as v dollars of value. Evaluate the change in mugs and the change in money *separately* using ϕ, then add. The reference point is current holdings; making no trade has value 0. Assume the person can afford all prices considered.

Buying. The person starts without a mug and can pay a price $p \geq 0$ to acquire one. Identify the gain and the loss. What is the largest price they will pay?

Selling. The person instead starts owning the mug and can receive p to give it up. Identify the gain and the loss. What is the smallest price they will accept?

At a price of \$10, would this person buy? Would they sell? What changes when $\lambda = 1$?

Present Bias

Suppose you have not had anything to drink for four hours and have just eaten salty potato chips. Which would you choose in each row?

Choice A	Choice B
1 sip now	2 sips in 5 minutes
1 sip in 20 minutes	2 sips in 25 minutes

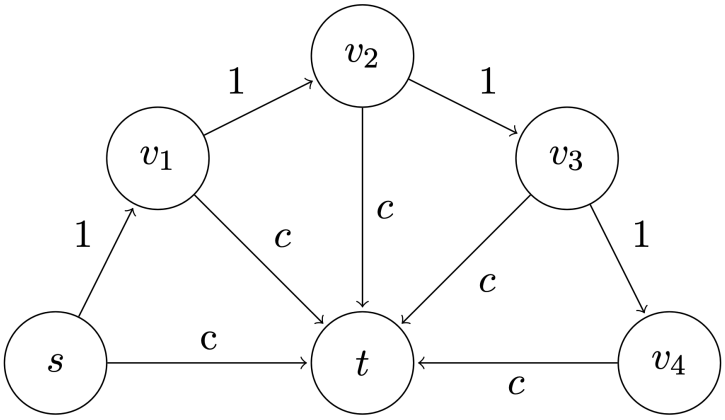
Both choices involve waiting five extra minutes for the second sip. Does delaying both options change your preference?

Sending a Package

George Akerlof described postponing mailing a package to Joseph Stiglitz. Sending the package has a fixed effort cost c, and each day of delay costs 1 in lost use. Sending after k days therefore has actual total cost $c+k$. An agent minimizing actual cost sends immediately.

A present-biased agent gives an immediate cost extra weight $b \geq 1$, while leaving future costs unchanged. They compare sending today with waiting one day and sending tomorrow. Only today's cost is multiplied by b.

Today versus tomorrow. What are the two perceived costs? When does the agent prefer waiting? Use $b = 2$ and $c = 3$ to compare the choices today, then make the comparison again tomorrow.



Here s is today, v_j is day j, and t means the package has been mailed. Each edge of cost 1 delays mailing by one day; each edge of cost c sends it. Suppose the package must be mailed by day 4.
Which route does the agent take for $b = 2, c = 3$? What is its actual total cost?

The graph rule. More generally, vertices describe progress toward a task and edges describe possible next actions. The graph is directed and acyclic, with start s, goal t, and nonnegative edge costs. Write $c(v, u)$ for the cost of taking an edge from v to u, and $d(u, t)$ for the ordinary cheapest remaining cost from u to the goal. At v, the perceived cost of taking that edge and then finishing is

$$b\,c(v, u) + d(u, t).$$

The agent chooses an outgoing edge minimizing this expression, then recomputes at the next vertex. They plan as if future choices minimize actual remaining cost; they do not anticipate their future re-planning. This is a *naive* present-biased agent. Fix a tie-breaking rule before any choices, and consider only vertices with a path to t.
What changes when $b = 1$?

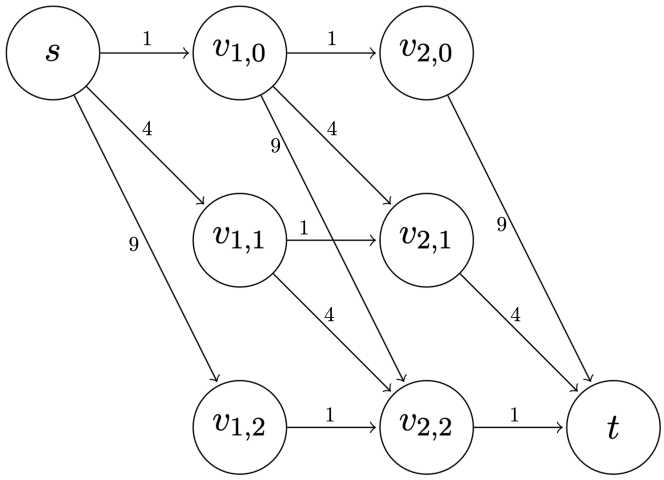
Deadlines and Commitment

Suppose students can choose binding submission deadlines for their assignments, up to the end of the course. A student who can always follow through on their plans can keep all options available by choosing the latest deadlines. What would you choose? Why?

Two Assignments in Three Weeks

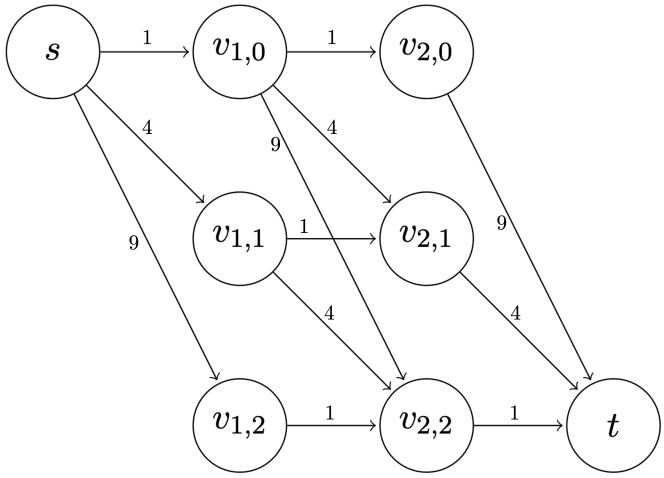
There are two assignments and three weeks. The effort cost in a week is 4 for completing one assignment, 9 for completing both, and 1 for completing none. The cost 1 applies even if both assignments are already done. In the graph, $v_{i,j}$ means that i weeks have passed and j assignments are complete; $s = v_{0,0}$ and $t = v_{3,2}$.

The student has $b = 2$ and receives a reward $R = 17$ only on reaching t. At each vertex, compare the least perceived remaining cost with R: continue if it is at most R, and otherwise abandon the course, receiving no reward. Past costs are already incurred and do not enter this comparison.



Without a deadline. At s, compare doing zero, one, or two assignments this week. What does the student plan to do later? Then recompute at each vertex they actually reach. Do they finish?

Can a deadline help? What deadline would you impose? Which vertex or edges would it remove? Recompute the student's choices in the retained graph, including the cheapest future costs. Does the student finish, and what actual total cost do they incur?



Why can fewer options help? Why can removing an option improve completion here? Could removing options lower the minimum actual cost of completing the course? Explain the difference.